Helpdesk Ticket Lite

Project Documentation

Project	Helpdesk Ticket Lite
Platform	Microsoft Power Platform: Dataverse, Power Apps, Power Automate, Power BI
Environment	Power Apps Developer Plan (developer environment)
Solution	HelpdeskLite (publisher prefix: hd)
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1. Problem Statement

Organisations need a simple way to track basic IT issues such as a laptop that won't boot or Wi-Fi that is down. Each issue should be recorded, routed to the right technician, and closed only after the user gives feedback on the fix.

2. Solution Overview

Helpdesk Ticket Lite is a small ticketing system. Employees raise tickets in a canvas app. Flows set deadlines, request approvals, collect feedback and flag late tickets. Power BI reports on the same data. Everything is stored in Dataverse and packaged in one solution named HelpdeskLite.

Layer	Tool	Role in the solution
Database	Dataverse	Stores the Ticket, Technician, Comment and FAQ tables and their relationships
App	Power Apps (canvas)	Helpdesk Lite: Browse, Details, Edit and Help screens
Automation	Power Automate	Five cloud flows for deadlines, approval, resolution, feedback and SLA checks
Reporting	Power BI Desktop	Volume by category and priority, average resolution time, status funnel
Assistant	Help screen (rule-based); Copilot Studio agent	The Help screen answers FAQs and raises tickets. The Copilot Studio agent was configured but could not be tested (no credits).
Packaging	Dataverse solution	HelpdeskLite holds the tables, the app and the flows

Ticket life cycle

1. A user creates a ticket from the Edit screen or the Help screen.
2. A flow sets the Due Date from the priority and adds a "Ticket received" comment. A Critical ticket also starts an approval request.
3. A coordinator assigns a technician and updates the status (Assigned, In Progress).
4. The technician sets the status to Resolved. A flow records Resolved On and adds a feedback request.
5. The user submits a 1 to 5 rating and a comment in the app.
6. A flow sets the status to Closed.
7. An hourly flow marks open tickets past their Due Date as SLA Breached.
8. Power BI reports on the same data.

Changes from the original brief

Brief	What was built	Reason
On create, post to a Teams "Helpdesk" channel	A comment is added to the ticket, with the Due Date set by the same flow	The practice account has no Teams licence
On Resolved, send a CSAT card	A feedback panel in the app, plus flows that record Resolved On and close the ticket	Same reason
Copilot Studio agent (stretch)	Agent configured but untested. A rule-based Help screen answers FAQs and creates tickets.	The tenant had no Copilot credits

3. Data Model

The data model has four tables. One technician can be assigned many tickets, and one ticket can have many comments. These links are stored as lookup columns, so a technician's name is kept once and every ticket points to it.

Ticket

Column	Type	Purpose
Title	Text (primary column)	Short name of the issue
Description	Multiple lines of text	Details of the problem
Category	Choice	Hardware, Network, Software, Access, Other
Priority	Choice	Low, Medium, High, Critical
Ticket Status	Choice	New, Assigned, In Progress, Resolved, Closed
Assigned To	Lookup to Technician	The technician handling the ticket
Due Date	Date and time	SLA deadline, set by a flow from the priority
Resolved On	Date and time	Set by a flow when the status becomes Resolved
Resolution Hours	Formula (number)	Hours from Created On to Resolved On; blank until resolved
CSAT Rating	Whole number	The user's rating, 1 to 5
CSAT Feedback	Text	The user's comment
SLA Breached	Yes/No	Set to Yes by the hourly checker when a ticket is late
Requester Email	Email	The person who raised the ticket
Created By, Created On	System columns	Built in; record who created the ticket and when

Technician, Comment and FAQ

Table	Columns	Purpose
Technician	Name (primary), Email, Specialty (Hardware, Network, Software, Access), Is Available (Yes/No)	The support staff that tickets are assigned to
Comment	Comment Text (primary), Ticket (lookup to Ticket)	Notes on a ticket, written by users or added by flows
FAQ	Keyword (primary), Answer (multiple lines of text)	Answers shown by the Help screen

Relationships

Relationship	Type	Lookup column	Meaning
Ticket to Technician	Many-to-one	Assigned To	Many tickets can be assigned to one technician; each ticket has one technician
Comment to Ticket	Many-to-one	Ticket (on Comment)	Many comments can belong to one ticket; each comment belongs to one ticket

Sample data

Three technicians (Ravi Kumar, Anita Nair, Joseph Mathew) and ten tickets were typed in by hand with mixed categories, priorities and statuses. More tickets were created through the app during testing. The due dates on the hand-typed tickets were chosen to test the overdue colour and the SLA checker, so they do not follow the priority rule.

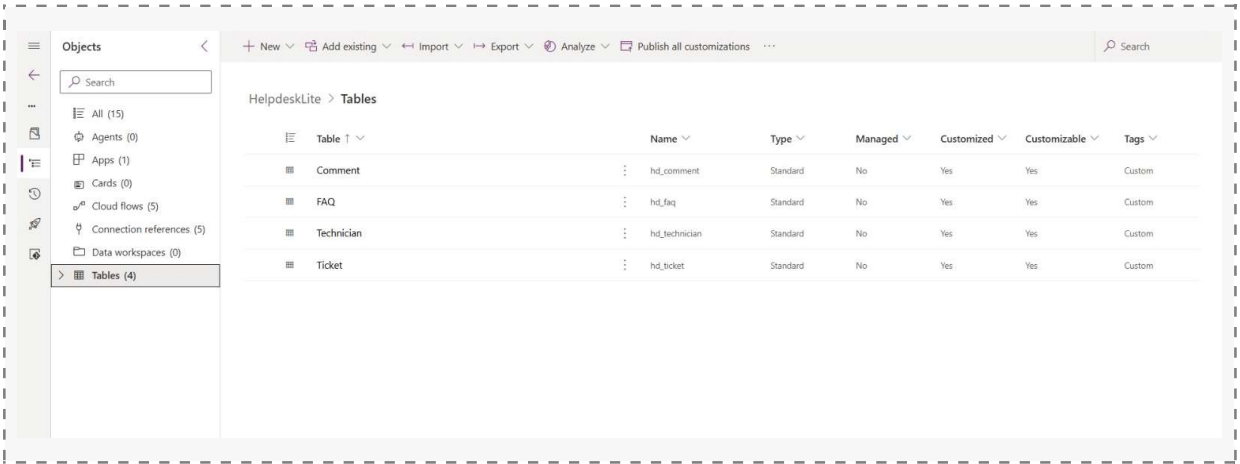


Figure 1: Solution HelpdeskLite showing its tables, app and cloud flows

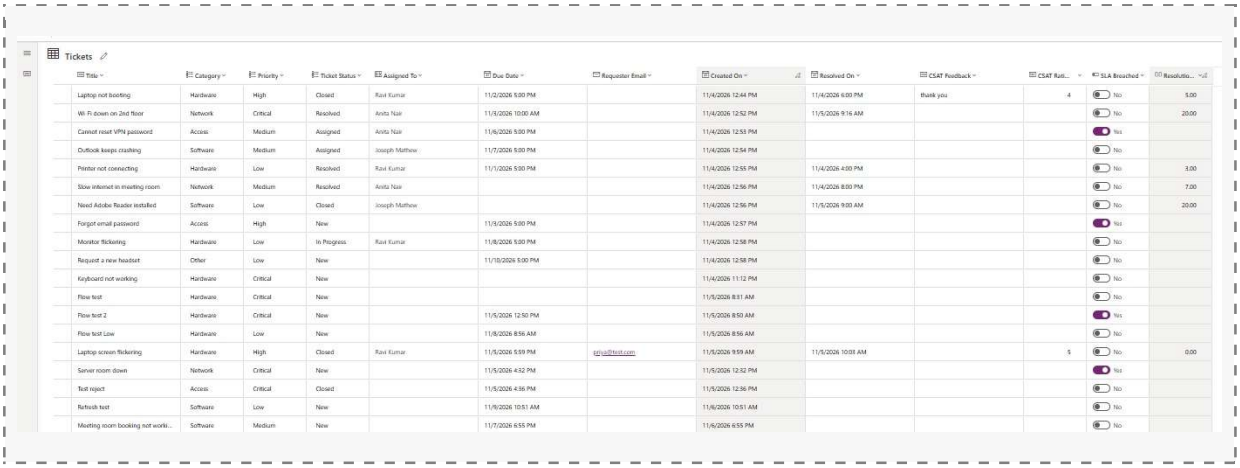


Figure 2: Ticket table columns

4. Canvas App

The canvas app Helpdesk Lite uses the tablet layout and four screens. Its data sources are the Tickets, Technicians, Comments and FAQs tables.

Screen	Purpose	Main controls
Browse (scrBrowse)	Find tickets and read their status at a glance	txtSearch, ddStatus, ddPriority, galTickets, btnNew, btnHelp
Details (scrDetails)	Read one ticket, its comments, add a comment, give feedback	frmDetails, galComments, txtNewComment, btnAddComment, feedback panel (ddRating, txtFeedback, btnSubmitFeedback), btnEdit, btnBack
Edit (scrEdit)	Create a new ticket or update an existing one	frmEdit, btnSave, btnCancel
Help (scrHelp)	Answer common questions and raise a ticket	txtQuestion, btnAsk, lblAnswer, txtTicketTitle, ddHelpCategory, ddHelpPriority, btnCreate

How the screens behave

- **Search and filters:** the ticket list shows only tickets whose title contains the search text and whose status and priority match the two dropdowns. "All" switches a filter off.
- **Details:** the form shows the selected ticket. The comments list shows only comments linked to that ticket, and Add comment saves a new Comment row linked to it.
- **Feedback panel:** it appears only when the ticket's status is Resolved and it has no rating yet. Submitting saves the rating and comment to the ticket, and a flow then closes it.
- **Edit:** + New ticket opens the form empty; Edit on the Details screen opens it with the ticket's values. Save writes the row, and Cancel discards the changes.
- **Help:** the assistant looks for an FAQ keyword in the question (for example vpn, wifi, laptop, printer, password) and shows that answer. If nothing matches it says so. A panel under the answer creates a ticket with a title, a category and a priority, with the status set to New.

Colours on the Browse screen

Element	Colour	Meaning
Status text	Blue	New
Status text	Purple	Assigned
Status text	Orange	In Progress
Status text	Green	Resolved
Status text	Grey	Closed
Due date	Red	The ticket is open and past its Due Date (SLA warning)
Due date	Grey	On time, or already Resolved or Closed
Row background	Light blue	The selected row

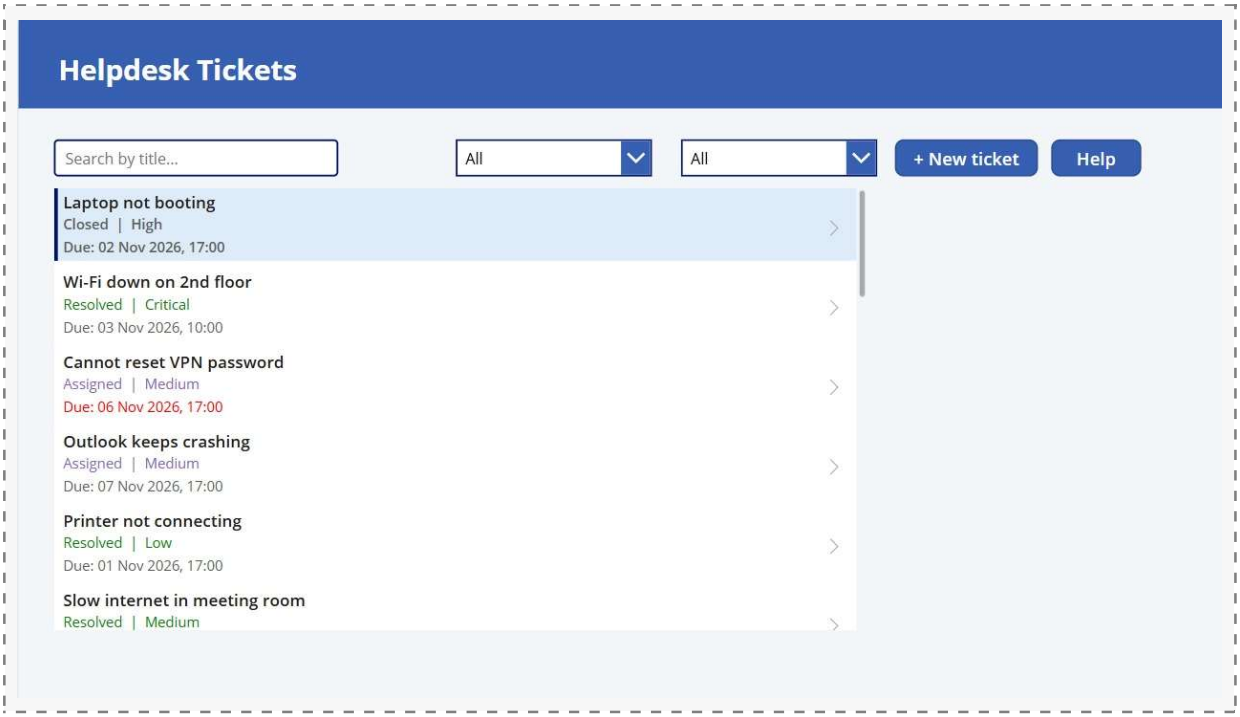


Figure 3: Browse screen with the search box, filters and colour-coded rows

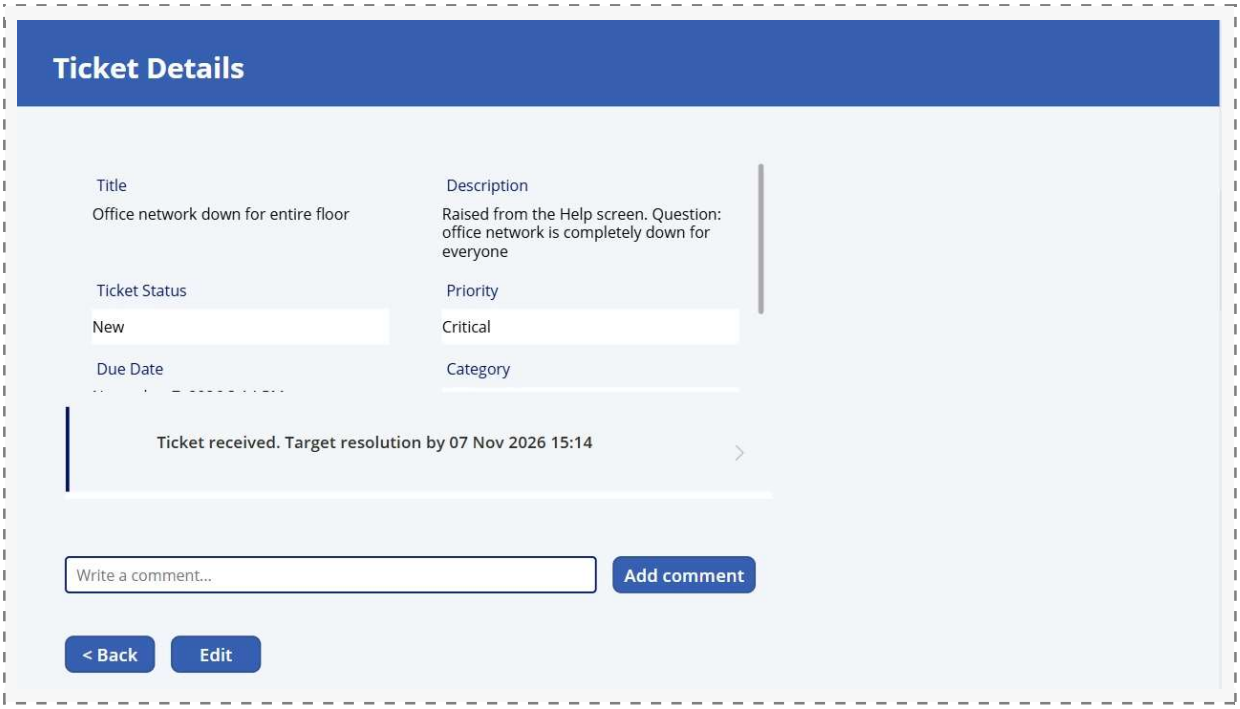


Figure 4: Details screen with comments

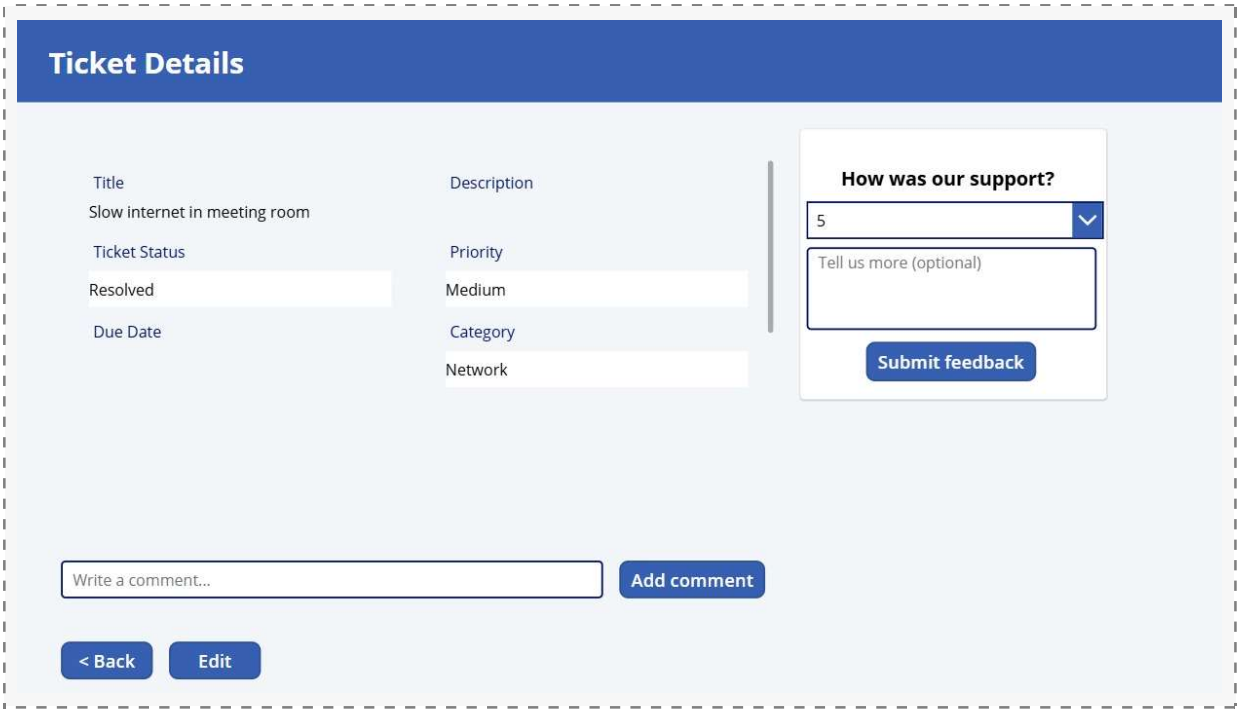


Figure 5: Details screen with the feedback panel for a Resolved ticket

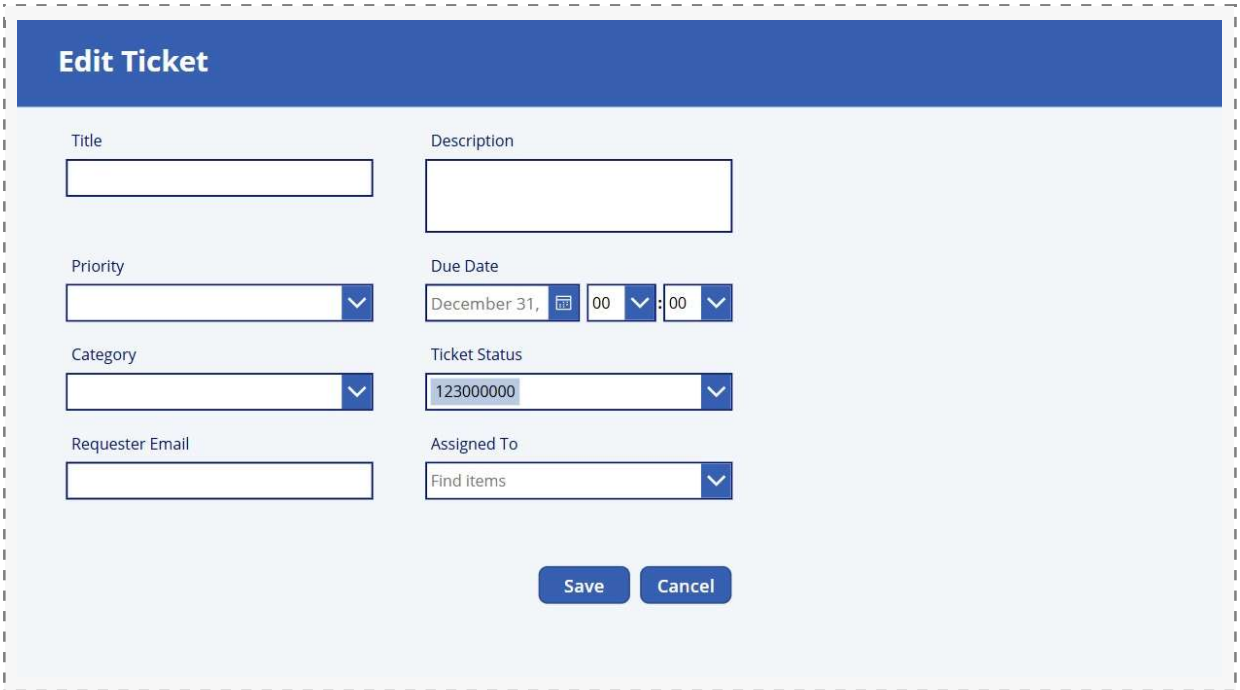


Figure 6: Edit screen

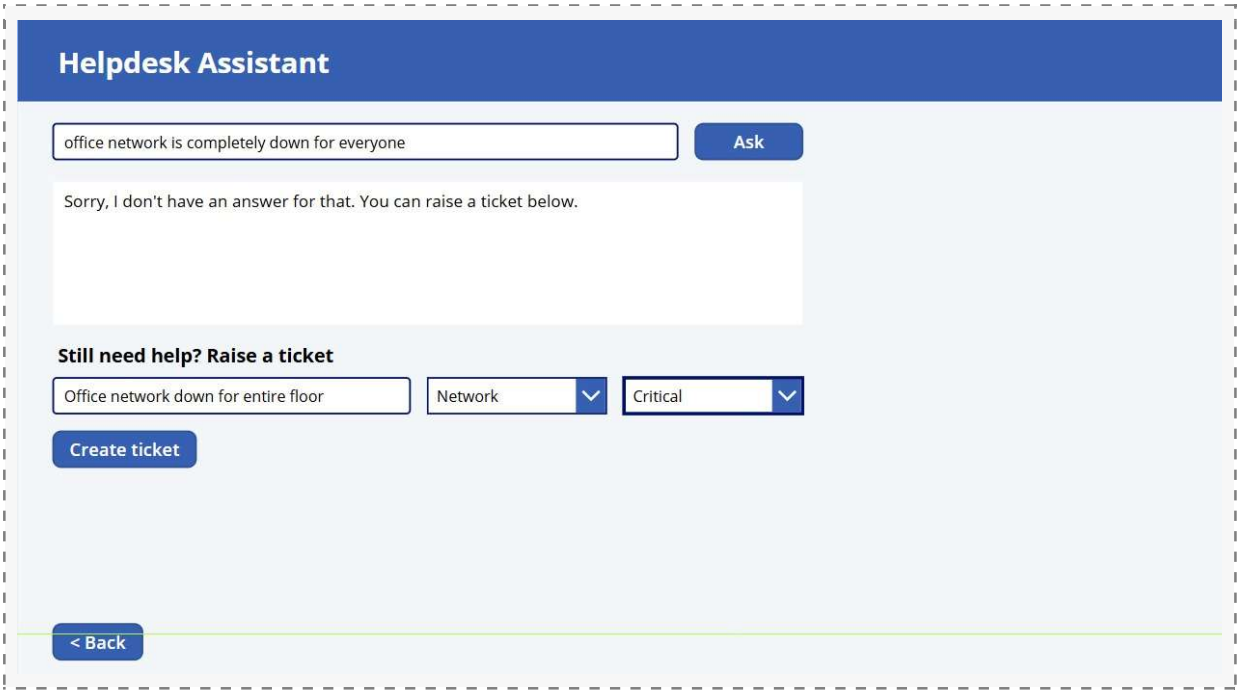


Figure 7: Help screen with an FAQ answer and the ticket panel

5. Automation

The automation is event driven. People change data in the app, and flows react to those changes without anyone clicking. One flow runs on a timer.

What starts each flow

Event	Trigger	Flow
A ticket is created	Dataverse: row added	On ticket created; Critical ticket approval
The status becomes Resolved	Dataverse: row modified (status column only, Resolved only)	On ticket resolved
A rating is saved	Dataverse: row modified (rating column only, not blank)	Close ticket on feedback
Every hour	Schedule	SLA breach checker

Avoiding loops. Flows that update the ticket they were triggered by would otherwise trigger themselves. The Modified triggers use Select columns and Filter rows so they run only when the one column they watch changes to the value they want.

SLA rules

Priority	Time allowed	Due Date
Critical	4 hours	Creation time plus 4 hours
High	8 hours	Creation time plus 8 hours
Medium	24 hours	Creation time plus 24 hours
Low	72 hours (default)	Creation time plus 72 hours

The target is the Due Date, set by a flow. The actual time is Resolution Hours, a formula column (Resolved On minus Created On). A ticket is breached when it is still open after its Due Date.

Status life cycle

Status	Set by	What happens next
New	The user, or the Help screen	A coordinator assigns a technician
Assigned	A coordinator in the Edit screen	The technician starts work
In Progress	The technician in the Edit screen	The technician resolves it
Resolved	The technician in the Edit screen	The flow stamps Resolved On; the feedback panel appears
Closed	The Close ticket on feedback flow (or the approval flow on a rejection)	End of the life cycle

Notifications

The practice account has no Teams or mailbox, so every flow records its message as a comment on the ticket. The comment appears in the ticket's comment list on the Details screen. In a real deployment these messages would be sent through Teams or email.

6. Power Automate Flows

The solution contains five cloud flows. Dataverse is a premium connector, which the Developer Plan covers. The flows run under the practice work account, and an approver answers requests on the Approvals page of make.powerautomate.com.

Flow	Trigger	Result
HD - On ticket created	Row added (Tickets)	Due Date set; "Ticket received" comment added
HD - On ticket resolved	Row modified (status)	Resolved On stamped; feedback request comment added
HD - Close ticket on feedback	Row modified (rating)	Status set to Closed
HD - Critical ticket approval	Row added (Critical only)	Manager approves or rejects; ticket commented or closed
HD - SLA breach checker	Every hour	Open overdue tickets set to SLA Breached = Yes

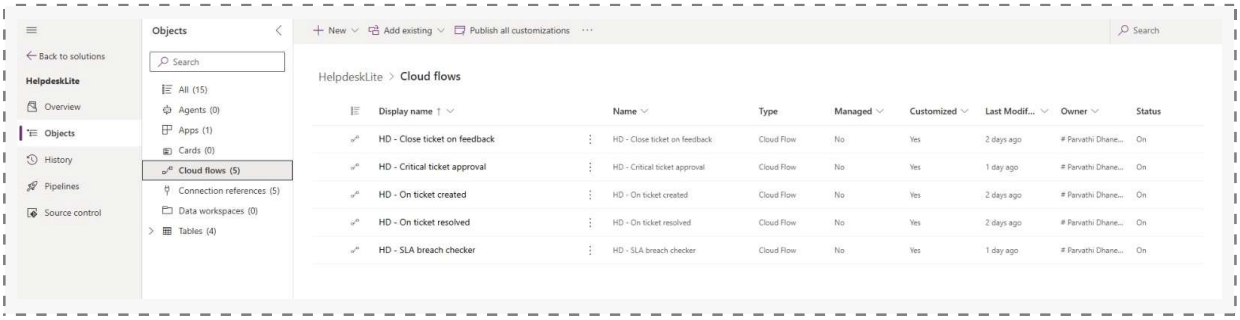


Figure 8: Solution cloud flows list showing all five flows

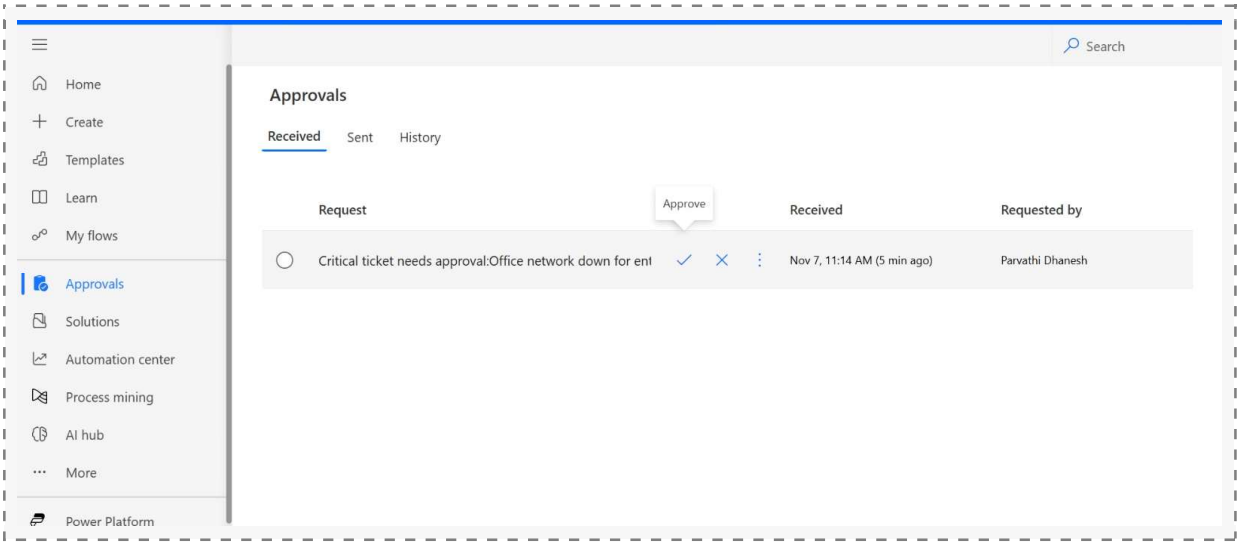


Figure 9: Approvals page showing a Critical ticket request

7. Power BI

The report is built in Power BI Desktop. It connects to Dataverse with the Dataverse connector, signs in with the work account, imports the Ticket and Technician tables, and is saved as HelpdeskLite.pbix.

Visuals

Visual	Type	Fields
Tickets by Category	Clustered bar chart	Category Label by count of Ticket
Tickets by Priority	Stacked column chart	Priority Label by count of Ticket
Avg Resolution Time (hours)	Card	Resolution Hours, summarised as Average
Ticket Status Funnel	Funnel	Status Label by count of Ticket

Example values at the time of the build

Visual	Values shown
By category	Hardware 8, Access 3, Network 3, Software 2, Other 1
By priority	Low 5, Medium 3, High 3, Critical 6
Average resolution time	9.17 hours
Status funnel	New 7, Closed 4, Resolved 3, Assigned 2, In Progress 1

These numbers change as tickets are added. Use Home, then Refresh in Power BI Desktop to load new data.

Notes

- The funnel counts tickets currently at each status, so it does not always narrow from top to bottom.
- The average resolution time uses only tickets that have a Resolved On value.
- The report is a local file. Publishing it online needs a Power BI account.

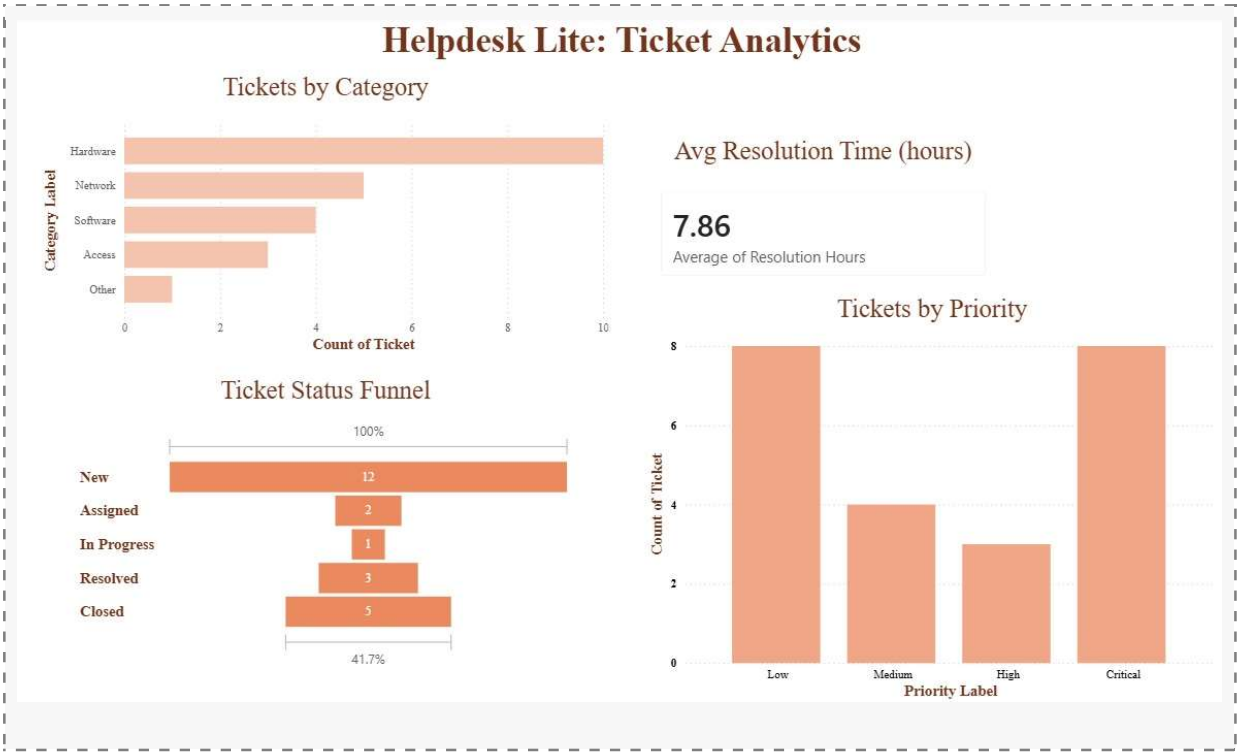


Figure 10: Power BI report with the four visuals

8. Acceptance Criteria

Main criterion: new ticket, assignment, resolution and feedback captured end to end. This was tested with a fresh ticket and passed.

#	Criterion	How it was verified	Result
1	A new ticket can be created	Created from + New ticket and from the Help screen; the row appears in the Ticket table	Passed
2	A deadline is set from the priority	New tickets received a Due Date about 4, 8, 24 or 72 hours ahead, plus a comment	Passed
3	A ticket can be assigned to a technician	Edit screen: Assigned To and status set; Details screen shows the technician	Passed
4	Tickets can be searched and filtered by status and priority	Search box and both dropdowns tested; counts matched the Power BI report	Passed
5	Resolution is recorded	Status set to Resolved; Resolved On filled; feedback request comment added	Passed
6	Feedback is captured and the ticket closes	Rating and comment saved to CSAT Rating and CSAT Feedback; status changed to Closed by the flow	Passed
7	Critical tickets need approval	One request approved and one rejected; comment added or ticket closed	Passed
8	Late tickets are flagged	Open overdue tickets set to SLA Breached = Yes; a resolved overdue ticket was left alone; red due dates in the app	Passed
9	Relationships exist	Relationships pages show Assigned To and Ticket as many-to-one	Passed

#	Criterion	How it was verified	Result
10	Analytics show volume and resolution time	Report shows volume by category and priority, average resolution time and a status funnel	Passed
11	Post to a Teams channel on create (brief)	Replaced by a comment on the ticket (no Teams licence)	Adapted
12	Send a CSAT card on Resolved (brief)	Replaced by the in-app feedback panel and flows	Adapted
13	Copilot Studio agent (stretch)	Agent configured, but not testable (0 Copilot credits). The Help screen substitute was tested.	Not tested; substitute passed

9. Limitations and Assumptions

Assumptions

- The project runs in one developer environment, created with the free Power Apps Developer Plan.
- The practice account has no Teams or mailbox, so notifications are comments on the ticket.
- One account plays every role (employee, coordinator, technician, approver).
- Dataverse is a premium connector and the Developer Plan covers its use.
- The sample tickets and technicians were typed in by hand.
- Times in comments are shown in India Standard Time.

Limitations

- The deadline and the approval are set when a ticket is created. Changing the priority later does not change the Due Date and does not start an approval, because no flow reacts to a priority change.
- The Help screen matches keywords. It is not AI, and a question without a known keyword gets a fallback message.
- The Copilot Studio agent was configured but could not be tested, because the tenant had no Copilot credits.
- The status funnel shows where tickets are now, so it does not always narrow.
- Everyone sees every ticket. No security roles were set up.
- Notifications are comments, not emails or Teams messages.
- The app shows changes made by a flow only after a screen is reopened or refreshed.
- The Power BI report is a local file and must be refreshed by hand.

10. Next Steps

1. **Handle priority changes.** Add a flow that triggers when Priority changes, recalculates the Due Date and starts an approval when the new priority is Critical.

2. **Send real notifications.** Post to a Teams channel and email the requester when a ticket is created, resolved or breached.
3. **Finish the Copilot Studio agent.** Add Copilot credits, test the FAQ answers, and add a tool that creates a ticket from chat.
4. **Add security roles.** Let employees see only their own tickets, and give technicians and coordinators wider access.
5. **Auto-assign tickets.** Pick a technician by Specialty and Is Available when a ticket is created.